

Informe de experimento: results_matrix_gpt-4o-mini

Generado el 2026-09-15 22:06 por benchmarks/experiment_report.py a partir de 56 report(s) del directorio results_matrix_gpt-4o-mini (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f
↪ low-agent/benchmarks/results_matrix_gpt-4o-mini
```

- Reports leídos: 56 (report.rescored.json usado en 56; el resto report.json)
- Modelos: none:rule_based, openrouter:openai/gpt-4o-mini
- Métodos: llm_only:cot, llm_only:few_shot, llm_only:rag, llm_only:structured, single_call:cot, single_call:few_shot, single_call:rag, single_call:structured, rule_based, react_nogate, react, plan_act_nogate, plan_act, pfagent
- Casos: case14, case30, case57, case118
- Requests: generadas, N=40 por caso y seed, difficulties=todas (56 reports)
- Items por dificultad: plain=560, parameterized=560, multistep=560, ambiguous=560
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-14 00:47 –2026-09-14 01:42
- Costo total: 5.0039 USD (2240 items; costo medio 0.00223 USD/item)
- Items totales: 2240; ok=2170; con error=70; tokens totales=32060315
- Solved global: 58.3% (1305/2240)

Errores por tipo:

Tipo	Items	Métodos afectados
formulation_failure	39	rule_based (39)
GroundTruthNotConverged	28	llm_only:cot (2), llm_only:few_shot (2), llm_only:rag (2), llm_only:structured (2), pfagent (2), plan_act (2), plan_act_nogate (2), react (2), react_nogate (2), rule_based (2), single_call:cot (2), single_call:few_shot (2), single_call:rag (2), single_call:structured (2)
no_solver_result	3	single_call:cot (3)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método llm_only:cot

System prompt –734 caracteres; strategy=cot, forced=False, sin herramientas

```
You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).

- Do not output NaN/Infinity; all values must be finite real numbers.

- If you cannot complete the calculation, set `converged` to false and output the structure you can provide (an empty array is acceptable).

Reasoning mode: you MAY write a short step-by-step reasoning in plain text before the answer. The final JSON object must be the LAST thing
↪ in the response, inside a single ```json fenced block, and it must be the only JSON object in the response. Do not use curly braces in
↪ the reasoning text.
```

User prompt (primer item) –6461 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

```
You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

## System Data (from pandapower network tables)

### Node Data (Bus Table: net.bus)
```

	name	vn_kv	type	zone	in_service
0	1	135.000	b	1.0	True
1	2	135.000	b	1.0	True
2	3	135.000	b	1.0	True
3	4	135.000	b	1.0	True
4	5	135.000	b	1.0	True
5	6	0.208	b	1.0	True
6	7	14.000	b	1.0	True
7	8	12.000	b	1.0	True
8	9	0.208	b	1.0	True
9	10	0.208	b	1.0	True
10	11	0.208	b	1.0	True

... (3 líneas omitidas de 15)

Load Data (Load Table: net.load)

	bus	p_mw	q_mvar	in_service
0	1	21.473798	12.567615	True
1	2	85.726702	17.290948	True
2	3	45.568126	-3.717902	True
3	4	6.887322	1.449962	True
4	5	11.740497	7.861940	True
5	8	32.229269	18.135792	True
6	9	9.714580	6.260507	True
7	10	3.424253	1.761044	True
8	11	6.539657	1.715320	True
9	12	14.757610	6.340306	True
10	13	15.783969	5.296634	True

Generator Data (Generator Table: net.gen)

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	1	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	2	0.00000	1.010	0.0	100.0	0.0	40.0	True
2	5	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	7	0.00000	1.090	0.0	100.0	-6.0	24.0	True

Balanced Node/External Grid (External Grid Table: net.ext_grid)

	bus	vm_pu	va_degree	in_service
0	0	1.06	0.0	True

Line Data (Line Table: net.line)

	from_bus	to_bus	length_km	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max_i_ka	df	parallel	in_service
0	0	1	1.0	3.532005	10.783732	768.484773	42.339020	1.0	1	True
1	0	4	1.0	9.846967	40.649040	716.088084	42.339020	1.0	1	True
2	1	2	1.0	8.563928	36.080033	637.493051	42.339020	1.0	1	True
3	1	3	1.0	10.590547	32.134320	494.857619	42.339020	1.0	1	True
4	1	4	1.0	10.379138	31.689630	503.590401	42.339020	1.0	1	True
5	2	3	1.0	12.212573	31.170217	186.299339	42.339020	1.0	1	True
6	3	4	1.0	2.433038	7.674548	0.000000	42.339020	1.0	1	True
7	5	10	1.0	0.000041	0.000086	0.000000	27479.652235	1.0	1	True
8	5	11	1.0	0.000053	0.000111	0.000000	27479.652235	1.0	1	True
9	5	12	1.0	0.000029	0.000056	0.000000	27479.652235	1.0	1	True
10	8	9	1.0	0.000014	0.000037	0.000000	27479.652235	1.0	1	True

... (4 líneas omitidas de 16)

Transformer Data (Transformer Table: net.trafo)

	hv_bus	lv_bus	sn_mva	vn_hv_kv	vn_lv_kv	vk_percent	vk_r_percent	pfe_kw	i0_percent	shift_degree	tap_side	tap_neutral	tap_min
↪	tap_max	tap_step_percent	in_service										
0	3	6	9900.0	135.0	14.000	2070.288	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN	2.2	True										
1	3	8	9900.0	135.0	0.208	5506.182	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN	3.1	True										
2	4	5	9900.0	135.0	0.208	2494.998	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN	6.8	True										
3	6	7	9900.0	14.0	12.000	1743.885	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN	NaN	True										
4	6	8	9900.0	14.0	0.208	1089.099	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN	NaN	True										

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Reasoning Instructions

Before producing the final JSON, reason step by step:

1. Which buses does the request refer to (display ids, slack bus, load and generator buses)?
 2. Which lines/branches are involved (from_bus-to_bus, line_id convention)?
 3. Which quantities and units are asked for (MW, Mvar, p.u., degrees, % loading)?
 4. What changes with respect to the base case, and which power balance must hold?
- Write the reasoning first, then the final JSON as the last part of the response.

Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
- 2) Active power (MW) and load factor (%) at the beginning of each branch
- For lines: `line\_id` uses the row index of `net.line`
- For transformers: `line\_id` uses the row index of 100000 + `net.trafo`
- 3) Total generation (MW), total load (MW), total losses (MW)
- 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

Método llm_only:few_shot

System prompt —439 caracteres; strategy=few_shot, forced=False, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- If you cannot complete the calculation, set `converged` to false and output the structure you can provide (an empty array is acceptable).

User prompt (primer item) —7383 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	vn_kv	type	zone	in_service
0	1	135.000	b	1.0	True
1	2	135.000	b	1.0	True
2	3	135.000	b	1.0	True
3	4	135.000	b	1.0	True
4	5	135.000	b	1.0	True
5	6	0.208	b	1.0	True
6	7	14.000	b	1.0	True
7	8	12.000	b	1.0	True
8	9	0.208	b	1.0	True
9	10	0.208	b	1.0	True
10	11	0.208	b	1.0	True

... (3 líneas omitidas de 15)

Load Data (Load Table: net.load)

	bus	p_mw	q_mvar	in_service
0	1	21.473798	12.567615	True
1	2	85.726702	17.290948	True
2	3	45.568126	-3.717902	True
3	4	6.887322	1.449962	True
4	5	11.740497	7.861940	True
5	8	32.229269	18.135792	True
6	9	9.714580	6.260507	True
7	10	3.424253	1.761044	True
8	11	6.539657	1.715320	True
9	12	14.757610	6.340306	True
10	13	15.783969	5.296634	True

Generator Data (Generator Table: net.gen)

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	1	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	2	0.00000	1.010	0.0	100.0	0.0	40.0	True

2	5	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	7	0.00000	1.090	0.0	100.0	-6.0	24.0	True

Balanced Node/External Grid (External Grid Table: net.ext_grid)

bus	vm_pu	va_degree	in_service
0	0	1.06	0.0
			True

Line Data (Line Table: net.line)

	from_bus	to_bus	length_km	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max_i_ka	df	parallel	in_service
0	0	1	1.0	3.532005	10.783732	768.484773	42.339020	1.0	1	True
1	0	4	1.0	9.846967	40.649040	716.088084	42.339020	1.0	1	True
2	1	2	1.0	8.563928	36.080033	637.493051	42.339020	1.0	1	True
3	1	3	1.0	10.590547	32.134320	494.857619	42.339020	1.0	1	True
4	1	4	1.0	10.379138	31.689630	503.590401	42.339020	1.0	1	True
5	2	3	1.0	12.212573	31.170217	186.299339	42.339020	1.0	1	True
6	3	4	1.0	2.433038	7.674548	0.000000	42.339020	1.0	1	True
7	5	10	1.0	0.000041	0.000086	0.000000	27479.652235	1.0	1	True
8	5	11	1.0	0.000053	0.000111	0.000000	27479.652235	1.0	1	True
9	5	12	1.0	0.000029	0.000056	0.000000	27479.652235	1.0	1	True
10	8	9	1.0	0.000014	0.000037	0.000000	27479.652235	1.0	1	True

... (4 líneas omitidas de 16)

Transformer Data (Transformer Table: net.trafo)

	hv_bus	lv_bus	sn_mva	vn_hv_kv	vn_lv_kv	vk_percent	vk_r_percent	pfe_kw	i0_percent	shift_degree	tap_side	tap_neutral	tap_min
	↪	tap_max	tap_step_percent	in_service									
0	3	6	9900.0	135.0	14.000	2070.288	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		2.2	True									
1	3	8	9900.0	135.0	0.208	5506.182	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		3.1	True									
2	4	5	9900.0	135.0	0.208	2494.998	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		6.8	True									
3	6	7	9900.0	14.0	12.000	1743.885	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN		NaN	True									
4	6	8	9900.0	14.0	0.208	1089.099	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN		NaN	True									

Worked Examples

The examples below use a FICTIONAL tiny 3-bus/2-line system that is NOT the case in this prompt; their numbers are illustrative only and must not be copied. They show the expected answer shape.

Example 1

Request: Run the AC power flow on this 3-bus example system and report all voltages and flows.

Expected answer (example numbers):

```
```json
{"converged": true,
 "bus_voltages": [{"bus_id": 1, "vm_pu": 1.05, "va_deg": 0.0},
 {"bus_id": 2, "vm_pu": 1.012, "va_deg": -2.4},
 {"bus_id": 3, "vm_pu": 0.987, "va_deg": -4.1}],
 "line_flows": [{"line_id": 0, "p_from_mw": 61.3, "loading_percent": 48.2},
 {"line_id": 1, "p_from_mw": 30.9, "loading_percent": 25.7}],
 "total_generation_mw": 91.8, "total_load_mw": 90.0, "total_loss_mw": 1.8}
```
```

Example 2

Request: In the same 3-bus example, set the load at bus 3 to 60 MW and report the new power flow.

Expected answer (example numbers):

```
```json
{"converged": true,
 "bus_voltages": [{"bus_id": 1, "vm_pu": 1.05, "va_deg": 0.0},
 {"bus_id": 2, "vm_pu": 1.004, "va_deg": -3.3},
 {"bus_id": 3, "vm_pu": 0.941, "va_deg": -6.2}],
 "line_flows": [{"line_id": 0, "p_from_mw": 82.9, "loading_percent": 66.0},
 {"line_id": 1, "p_from_mw": 40.6, "loading_percent": 34.1}],
 "total_generation_mw": 123.4, "total_load_mw": 120.0, "total_loss_mw": 3.4}
```
```

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
- 2) Active power (MW) and load factor (%) at the beginning of each branch
 - For lines: `line\_id` uses the row index of `net.line`
 - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`
- 3) Total generation (MW), total load (MW), total losses (MW)
- 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

Método llm_only:rag

System prompt —439 caracteres; strategy=rag, forced=False, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- If you cannot complete the calculation, set `converged` to false and output the structure you can provide (an empty array is acceptable).

User prompt (primer item) —1882 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (retrieved rows relevant to the request; 8 of 34 rows)

Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
↪ index, transformers use 100000+row index.

Buses

```
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
```

Branches

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
 - 2) Active power (MW) and load factor (%) at the beginning of each branch
- For lines: `line\_id` uses the row index of `net.line`
 - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`
- 3) Total generation (MW), total load (MW), total losses (MW)
 - 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

Método llm_only:structured

System prompt —439 caracteres; strategy=structured, forced=False, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- If you cannot complete the calculation, set `converged` to false and output the structure you can provide (an empty array is acceptable).

User prompt (primer item) —5966 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

| | name | vn_kv | type | zone | in_service |
|----|------|---------|------|------|------------|
| 0 | 1 | 135.000 | b | 1.0 | True |
| 1 | 2 | 135.000 | b | 1.0 | True |
| 2 | 3 | 135.000 | b | 1.0 | True |
| 3 | 4 | 135.000 | b | 1.0 | True |
| 4 | 5 | 135.000 | b | 1.0 | True |
| 5 | 6 | 0.208 | b | 1.0 | True |
| 6 | 7 | 14.000 | b | 1.0 | True |
| 7 | 8 | 12.000 | b | 1.0 | True |
| 8 | 9 | 0.208 | b | 1.0 | True |
| 9 | 10 | 0.208 | b | 1.0 | True |
| 10 | 11 | 0.208 | b | 1.0 | True |

... (3 líneas omitidas de 15)

Load Data (Load Table: net.load)

| | bus | p_mw | q_mvar | in_service |
|----|-----|-----------|-----------|------------|
| 0 | 1 | 21.473798 | 12.567615 | True |
| 1 | 2 | 85.726702 | 17.290948 | True |
| 2 | 3 | 45.568126 | -3.717902 | True |
| 3 | 4 | 6.887322 | 1.449962 | True |
| 4 | 5 | 11.740497 | 7.861940 | True |
| 5 | 8 | 32.229269 | 18.135792 | True |
| 6 | 9 | 9.714580 | 6.260507 | True |
| 7 | 10 | 3.424253 | 1.761044 | True |
| 8 | 11 | 6.539657 | 1.715320 | True |
| 9 | 12 | 14.757610 | 6.340306 | True |
| 10 | 13 | 15.783969 | 5.296634 | True |

Generator Data (Generator Table: net.gen)

| | bus | p_mw | vm_pu | min_p_mw | max_p_mw | min_q_mvar | max_q_mvar | in_service |
|---|-----|----------|-------|----------|----------|------------|------------|------------|
| 0 | 1 | 41.63323 | 1.045 | 0.0 | 140.0 | -40.0 | 50.0 | True |
| 1 | 2 | 0.00000 | 1.010 | 0.0 | 100.0 | 0.0 | 40.0 | True |
| 2 | 5 | 0.00000 | 1.070 | 0.0 | 100.0 | -6.0 | 24.0 | True |
| 3 | 7 | 0.00000 | 1.090 | 0.0 | 100.0 | -6.0 | 24.0 | True |

Balanced Node/External Grid (External Grid Table: net.ext_grid)

| | bus | vm_pu | va_degree | in_service |
|---|-----|-------|-----------|------------|
| 0 | 0 | 1.06 | 0.0 | True |

Line Data (Line Table: net.line)

| | from_bus | to_bus | length_km | r_ohm_per_km | x_ohm_per_km | c_nf_per_km | max_i_ka | df | parallel | in_service |
|----|----------|--------|-----------|--------------|--------------|-------------|--------------|-----|----------|------------|
| 0 | 0 | 1 | 1.0 | 3.532005 | 10.783732 | 768.484773 | 42.339020 | 1.0 | 1 | True |
| 1 | 0 | 4 | 1.0 | 9.846967 | 40.649040 | 716.088084 | 42.339020 | 1.0 | 1 | True |
| 2 | 1 | 2 | 1.0 | 8.563928 | 36.080033 | 637.493051 | 42.339020 | 1.0 | 1 | True |
| 3 | 1 | 3 | 1.0 | 10.590547 | 32.134320 | 494.857619 | 42.339020 | 1.0 | 1 | True |
| 4 | 1 | 4 | 1.0 | 10.379138 | 31.689630 | 503.590401 | 42.339020 | 1.0 | 1 | True |
| 5 | 2 | 3 | 1.0 | 12.212573 | 31.170217 | 186.299339 | 42.339020 | 1.0 | 1 | True |
| 6 | 3 | 4 | 1.0 | 2.433038 | 7.674548 | 0.000000 | 42.339020 | 1.0 | 1 | True |
| 7 | 5 | 10 | 1.0 | 0.000041 | 0.000086 | 0.000000 | 27479.652235 | 1.0 | 1 | True |
| 8 | 5 | 11 | 1.0 | 0.000053 | 0.000111 | 0.000000 | 27479.652235 | 1.0 | 1 | True |
| 9 | 5 | 12 | 1.0 | 0.000029 | 0.000056 | 0.000000 | 27479.652235 | 1.0 | 1 | True |
| 10 | 8 | 9 | 1.0 | 0.000014 | 0.000037 | 0.000000 | 27479.652235 | 1.0 | 1 | True |

... (4 líneas omitidas de 16)

Transformer Data (Transformer Table: net.trafo)

| | hv_bus | lv_bus | sn_mva | vn_hv_kv | vn_lv_kv | vk_percent | vkr_percent | pfe_kw | i0_percent | shift_degree | tap_side | tap_neutral | tap_min |
|---|---------|------------------|------------|----------|----------|------------|-------------|--------|------------|--------------|----------|-------------|---------|
| ↪ | tap_max | tap_step_percent | in_service | | | | | | | | | | |
| 0 | 3 | 6 | 9900.0 | 135.0 | 14.000 | 2070.288 | 0.0 | 0.0 | 0.0 | 0.0 | hv | 0.0 | NaN |
| ↪ | NaN | 2.2 | True | | | | | | | | | | |

| | | | | | | | | | | | | | |
|---|-----|---|--------|-------|--------|----------|-----|-----|-----|-----|------|-----|-----|
| 1 | 3 | 8 | 9900.0 | 135.0 | 0.208 | 5506.182 | 0.0 | 0.0 | 0.0 | 0.0 | hv | 0.0 | NaN |
| ↪ | NaN | | 3.1 | True | | | | | | | | | |
| 2 | 4 | 5 | 9900.0 | 135.0 | 0.208 | 2494.998 | 0.0 | 0.0 | 0.0 | 0.0 | hv | 0.0 | NaN |
| ↪ | NaN | | 6.8 | True | | | | | | | | | |
| 3 | 6 | 7 | 9900.0 | 14.0 | 12.000 | 1743.885 | 0.0 | 0.0 | 0.0 | 0.0 | None | NaN | NaN |
| ↪ | NaN | | NaN | True | | | | | | | | | |
| 4 | 6 | 8 | 9900.0 | 14.0 | 0.208 | 1089.099 | 0.0 | 0.0 | 0.0 | 0.0 | None | NaN | NaN |
| ↪ | NaN | | NaN | True | | | | | | | | | |

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
- 2) Active power (MW) and load factor (%) at the beginning of each branch
 - For lines: `line\_id` uses the row index of `net.line`
 - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`
- 3) Total generation (MW), total load (MW), total losses (MW)
- 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

Método single_call:cot

System prompt —1697 caracteres; strategy=cot; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

What you can do

You complete tasks by calling tool functions:

1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

Key rules

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format

After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

Single-call mode

You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns, and do not fabricate numbers.

User prompt (primer item) —4192 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

System Data (compact, full case)

Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
↪ index, transformers use 100000+row index.

```
### Buses
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 7 | 14 kV
bus 8 | 12 kV | gen 0 MW vm 1.09 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
bus 11 | 0.208 kV | load 3.424 MW 1.761 Mvar
bus 12 | 0.208 kV | load 6.54 MW 1.715 Mvar
... (2 líneas omitidas de 14)

### Branches
line 0 | 1-2 | r 3.532 x 10.78 ohm/km | len 1 km | max_i 42.34 kA
line 1 | 1-5 | r 9.847 x 40.65 ohm/km | len 1 km | max_i 42.34 kA
line 2 | 2-3 | r 8.564 x 36.08 ohm/km | len 1 km | max_i 42.34 kA
line 3 | 2-4 | r 10.59 x 32.13 ohm/km | len 1 km | max_i 42.34 kA
line 4 | 2-5 | r 10.38 x 31.69 ohm/km | len 1 km | max_i 42.34 kA
line 5 | 3-4 | r 12.21 x 31.17 ohm/km | len 1 km | max_i 42.34 kA
line 6 | 4-5 | r 2.433 x 7.675 ohm/km | len 1 km | max_i 42.34 kA
line 7 | 6-11 | r 4.109e-05 x 8.605e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 8 | 6-12 | r 5.318e-05 x 0.0001107 ohm/km | len 1 km | max_i 2.748e+04 kA
line 9 | 6-13 | r 2.862e-05 x 5.636e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 10 | 9-10 | r 1.376e-05 x 3.656e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 11 | 9-14 | r 5.499e-05 x 0.000117 ohm/km | len 1 km | max_i 2.748e+04 kA
... (8 líneas omitidas de 20)
```

```
### Available tools
- load_case(case_name): Load IEEE test case.
- run_powerflow(no arguments): Run power flow on current case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(no arguments): Get most loaded branch.
- run_n1_contingency(top_k, criteria, max_candidates): Run N-1 contingency analysis.
- recommend_remedial_actions(max_actions, allow_load_shed, allow_voltage_control, include_best_comparison): Recommend remedial actions.
- apply_remedial_action(action_index, confirmed): Apply one remedial action by index.
- get_status(no arguments): Get current network status.
- generate_plot(plot_type): Generate one plot.
```

```
## Task
Load case14 and disconnect the line between bus ten and bus eleven.
```

```
## Reasoning Instructions
Before emitting tool calls, reason step by step and write a SHORT numbered plan in the message content:
1. Which buses (display ids), lines/branches and quantities/units the request refers to.
2. Which tools are needed and in which order, with the exact argument values.
Then, in the same response, emit the tool calls that implement the plan.
```

```
## Output Requirements
Answer by emitting tool calls (function calling) in THIS single response:
- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.
```

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
↪ non-converged result, say so.

Método single_call:few_shot

System prompt —1697 caracteres; strategy=few_shot; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
```

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format

After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

Single-call mode

You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to

- call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns,
- and do not fabricate numbers.

User prompt (primer item) —4475 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

System Data (compact, full case)

Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
→ index, transformers use 100000+row index.

Buses

```
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 7 | 14 kV
bus 8 | 12 kV | gen 0 MW vm 1.09 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
bus 11 | 0.208 kV | load 3.424 MW 1.761 Mvar
bus 12 | 0.208 kV | load 6.54 MW 1.715 Mvar
... (2 líneas omitidas de 14)
```

Branches

```
line 0 | 1-2 | r 3.532 x 10.78 ohm/km | len 1 km | max_i 42.34 kA
line 1 | 1-5 | r 9.847 x 40.65 ohm/km | len 1 km | max_i 42.34 kA
line 2 | 2-3 | r 8.564 x 36.08 ohm/km | len 1 km | max_i 42.34 kA
line 3 | 2-4 | r 10.59 x 32.13 ohm/km | len 1 km | max_i 42.34 kA
line 4 | 2-5 | r 10.38 x 31.69 ohm/km | len 1 km | max_i 42.34 kA
line 5 | 3-4 | r 12.21 x 31.17 ohm/km | len 1 km | max_i 42.34 kA
line 6 | 4-5 | r 2.433 x 7.675 ohm/km | len 1 km | max_i 42.34 kA
line 7 | 6-11 | r 4.109e-05 x 8.605e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 8 | 6-12 | r 5.318e-05 x 0.0001107 ohm/km | len 1 km | max_i 2.748e+04 kA
line 9 | 6-13 | r 2.862e-05 x 5.636e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 10 | 9-10 | r 1.376e-05 x 3.656e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 11 | 9-14 | r 5.499e-05 x 0.000117 ohm/km | len 1 km | max_i 2.748e+04 kA
... (8 líneas omitidas de 20)
```

Available tools

- load_case(case_name): Load IEEE test case.
- run_powerflow(no arguments): Run power flow on current case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(no arguments): Get most loaded branch.
- run_n1_contingency(top_k, criteria, max_candidates): Run N-1 contingency analysis.
- recommend_remedial_actions(max_actions, allow_load_shed, allow_voltage_control, include_best_comparison): Recommend remedial actions.
- apply_remedial_action(action_index, confirmed): Apply one remedial action by index.
- get_status(no arguments): Get current network status.
- generate_plot(plot_type): Generate one plot.

Worked Examples

The examples below refer to a DIFFERENT, fictional small system, not the case in this prompt; they show which tools to call, with which
→ arguments and in which order.

Example 1

Request: Disconnect the line between bus 2 and bus 4 and tell me which branch is now the most loaded.

Correct tool calls, in order:

1. disconnect_line(from_bus=2, to_bus=4)
2. get_most_loaded_branch()

Example 2

Request: Set the load at bus 5 to 40 MW and 10 Mvar, then rank the 3 worst N-1 contingencies by overload.

Correct tool calls, in order:

1. modify_load(bus_id=5, p_mw=40, q_mvar=10)
2. run_n1_contingency(top_k=3, criteria="max_overload")

```
## Task
Load case14 and disconnect the line between bus ten and bus eleven.

## Output Requirements
Answer by emitting tool calls (function calling) in THIS single response:
- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.
```

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
↳ non-converged result, say so.

Método single_call:rag

System prompt —1697 caracteres; strategy=rag; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

## Single-call mode
You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to
↳ call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns,
↳ and do not fabricate numbers.
```

User prompt (primer item) —1699 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

```
## System Data (retrieved rows relevant to the request; 8 of 34 rows)
Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
↳ index, transformers use 100000+row index.
```

```
### Buses
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
```

```
### Branches
```

```
### Available tools
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- load_case(case_name): Load IEEE test case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
```

```
## Task
Load case14 and disconnect the line between bus ten and bus eleven.
```

```
## Output Requirements
Answer by emitting tool calls (function calling) in THIS single response:
```

- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
 ↪ non-converged result, say so.

Método single_call:structured

System prompt —1697 caracteres; strategy=structured; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

What you can do

You complete tasks by calling tool functions:

1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

Key rules

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format

After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

Single-call mode

You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to

- ↪ call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns,
- ↪ and do not fabricate numbers.

User prompt (primer item) —3821 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

System Data (compact, full case)

Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
 ↪ index, transformers use 100000+row index.

Buses

```
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 7 | 14 kV
bus 8 | 12 kV | gen 0 MW vm 1.09 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
bus 11 | 0.208 kV | load 3.424 MW 1.761 Mvar
bus 12 | 0.208 kV | load 6.54 MW 1.715 Mvar
... (2 líneas omitidas de 14)
```

Branches

```
line 0 | 1-2 | r 3.532 x 10.78 ohm/km | len 1 km | max_i 42.34 kA
line 1 | 1-5 | r 9.847 x 40.65 ohm/km | len 1 km | max_i 42.34 kA
line 2 | 2-3 | r 8.564 x 36.08 ohm/km | len 1 km | max_i 42.34 kA
line 3 | 2-4 | r 10.59 x 32.13 ohm/km | len 1 km | max_i 42.34 kA
line 4 | 2-5 | r 10.38 x 31.69 ohm/km | len 1 km | max_i 42.34 kA
line 5 | 3-4 | r 12.21 x 31.17 ohm/km | len 1 km | max_i 42.34 kA
line 6 | 4-5 | r 2.433 x 7.675 ohm/km | len 1 km | max_i 42.34 kA
line 7 | 6-11 | r 4.109e-05 x 8.605e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 8 | 6-12 | r 5.318e-05 x 0.0001107 ohm/km | len 1 km | max_i 2.748e+04 kA
line 9 | 6-13 | r 2.862e-05 x 5.636e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 10 | 9-10 | r 1.376e-05 x 3.656e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 11 | 9-14 | r 5.499e-05 x 0.000117 ohm/km | len 1 km | max_i 2.748e+04 kA
```

... (8 líneas omitidas de 20)

```
### Available tools
- load_case(case_name): Load IEEE test case.
- run_powerflow(no arguments): Run power flow on current case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(no arguments): Get most loaded branch.
- run_n1_contingency(top_k, criteria, max_candidates): Run N-1 contingency analysis.
- recommend_remedial_actions(max_actions, allow_load_shed, allow_voltage_control, include_best_comparison): Recommend remedial actions.
- apply_remedial_action(action_index, confirmed): Apply one remedial action by index.
- get_status(no arguments): Get current network status.
- generate_plot(plot_type): Generate one plot.
```

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Answer by emitting tool calls (function calling) in THIS single response:

- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
↪ non-converged result, say so.

Método rule_based

Sin prompt —67 caracteres; rule_based no usa LLM: la request se analiza con un parser de expresiones regulares

Load case14 and disconnect the line between bus ten and bus eleven.

Método react_nogate

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

What you can do

You complete tasks by calling tool functions:

1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

Key rules

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format

After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

User prompt (primer item) = request_text —67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

Método react

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=True, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

What you can do

You complete tasks by calling tool functions:

1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

Key rules

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format

After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

User prompt (primer item) = request_text −67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

Método plan_act_nogate

System prompt (SYSTEM_PROMPT_EN) −1375 caracteres; architecture=plan_act, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

What you can do

You complete tasks by calling tool functions:

1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

Key rules

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format

After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

User prompt (primer item) = request_text −67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

PLAN_SYSTEM_PROMPT (cabecera) −1043 caracteres; primeras 12 líneas de 1424 caracteres; catálogo de tools omitido

You are the planner of a power-system analysis agent. You cannot call tools yourself. Given the user's request, output ONLY a JSON object of the form

```
{ "plan": [{"tool": "<tool_name>", "args": {...}, ...}]
```

listing, in execution order, every tool call needed to answer the request. Use only the tools below with exactly these argument names (* = required). Do not include explanations, markdown, or any text outside the JSON.

Available tools:

- load_case(case_name: string*): Load IEEE test case.
- run_powerflow(none): Run power flow on current case.
- modify_load(bus_id: integer*, p_mw: number*, q_mvar: number): Modify bus load and recompute.
- disconnect_line(from_bus: integer*, to_bus: integer*): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus: integer*, to_bus: integer*): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(none): Get most loaded branch.
- run_n1_contingency(top_k: integer, criteria: string, max_candidates: integer): Run N-1 contingency analysis.

...

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade tras ejecutar el plan

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
→ non-converged result, say so.

Método plan_act

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=plan_act, gate=True, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text —67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

PLAN_SYSTEM_PROMPT (cabecera) —1043 caracteres; primeras 12 líneas de 1424 caracteres; catálogo de tools omitido

You are the planner of a power-system analysis agent. You cannot call tools yourself. Given the user's request, output ONLY a JSON object
→ of the form

```
{ "plan": [{"tool": "<tool_name>", "args": {...}], ...}
```

listing, in execution order, every tool call needed to answer the request. Use only the tools below with exactly these argument names (* =
→ required). Do not include explanations, markdown, or any text outside the JSON.

```
Available tools:
- load_case(case_name: string*): Load IEEE test case.
- run_powerflow((none)): Run power flow on current case.
- modify_load(bus_id: integer*, p_mw: number*, q_mvar: number): Modify bus load and recompute.
- disconnect_line(from_bus: integer*, to_bus: integer*): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus: integer*, to_bus: integer*): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch((none)): Get most loaded branch.
- run_n1_contingency(top_k: integer, criteria: string, max_candidates: integer): Run N-1 contingency analysis.
...
```

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade tras ejecutar el plan

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
→ non-converged result, say so.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=True, memory=True

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
```

- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format

After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

User prompt (primer item) = request_text -67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del **scoreboard_per_case**): Solved = respondió correctamente de extremo a extremo; Form. = formulación exacta de las tool calls; V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado; FR = tasa de convergencia/feasibility; Faith. = números de la respuesta trazables a salidas de tools; SFR = fallos declarados con seguridad; Claim = éxito reclamado sobre un fallo; Abst. = abstención en items resolubles; Calls = tool calls medias; Tok. = tokens medios; TTV = tiempo medio (s); \$ = costo total USD. Porcentajes en %.

Por método $\times$ caso

Modelos: M1 = none:rule_based; M2 = openrouter:openai/gpt-4o-mini

| Model | Method | Case | N | Solved | Form. | V_MAE | F_MAE | FR | Faith. | SFR | Claim | Abst. | Calls | Tok. | TTV | \$ |
|-------|------------------------|---------|----|--------|-------|----------|-------|-------|--------|-----|-------|-------|-------|-------|-----|--------|
| M1 | rule_based | case14 | 40 | 72.5 | 72.5 | 1.89e-05 | 0.06 | 100.0 | 100.0 | - | - | 22.5 | 1.9 | n/a | 0.2 | 0.0000 |
| M1 | rule_based | case30 | 40 | 67.5 | 67.5 | 4.82e-05 | 0.06 | 100.0 | 100.0 | - | - | 27.5 | 1.9 | n/a | 0.1 | 0.0000 |
| M1 | rule_based | case57 | 40 | 62.5 | 62.5 | 4.09e-05 | 0.12 | 100.0 | 100.0 | 0.0 | 0.0 | 28.9 | 1.7 | n/a | 0.2 | 0.0000 |
| M1 | rule_based | case118 | 40 | 67.5 | 67.5 | 4.49e-06 | 1.47 | 100.0 | 100.0 | - | - | 27.5 | 1.7 | n/a | 0.1 | 0.0000 |
| M2 | llm_only:cot | case14 | 40 | 0.0 | - | - | - | 0.0 | 11.7 | - | - | 100.0 | n/a | 3046 | 5.1 | 0.0264 |
| M2 | llm_only:cot | case30 | 40 | 0.0 | - | - | - | 0.0 | 15.0 | - | - | 100.0 | n/a | 4297 | 4.9 | 0.0329 |
| M2 | llm_only:cot | case57 | 40 | 0.0 | - | - | - | 0.0 | 15.6 | 0.0 | 0.0 | 100.0 | n/a | 7611 | 3.6 | 0.0509 |
| M2 | llm_only:cot | case118 | 40 | 0.0 | - | - | - | 0.0 | 18.2 | - | - | 100.0 | n/a | 17062 | 9.1 | 0.1149 |
| M2 | llm_only:few_shot | case14 | 40 | 0.0 | - | - | - | 0.0 | 5.0 | - | - | 100.0 | n/a | 2984 | 1.9 | 0.0190 |
| M2 | llm_only:few_shot | case30 | 40 | 0.0 | - | - | - | 0.0 | 7.5 | - | - | 100.0 | n/a | 4290 | 2.2 | 0.0269 |
| M2 | llm_only:few_shot | case57 | 40 | 0.0 | - | - | - | 0.0 | 5.3 | 0.0 | 0.0 | 100.0 | n/a | 7560 | 1.2 | 0.0441 |
| M2 | llm_only:few_shot | case118 | 40 | 0.0 | - | - | - | 0.0 | 7.5 | - | - | 100.0 | n/a | 16758 | 2.5 | 0.1017 |
| M2 | llm_only:rag | case14 | 40 | 0.0 | - | - | - | 0.0 | 5.0 | - | - | 100.0 | n/a | 845 | 1.8 | 0.0062 |
| M2 | llm_only:rag | case30 | 40 | 0.0 | - | - | - | 0.0 | 7.5 | - | - | 100.0 | n/a | 818 | 1.7 | 0.0060 |
| M2 | llm_only:rag | case57 | 40 | 0.0 | - | - | - | 0.0 | 5.3 | 0.0 | 0.0 | 100.0 | n/a | 856 | 1.9 | 0.0059 |
| M2 | llm_only:rag | case118 | 40 | 0.0 | - | - | - | 0.0 | 7.5 | - | - | 100.0 | n/a | 844 | 1.7 | 0.0062 |
| M2 | llm_only:structured | case14 | 40 | 0.0 | - | - | - | 0.0 | 5.0 | - | - | 100.0 | n/a | 2480 | 1.8 | 0.0160 |
| M2 | llm_only:structured | case30 | 40 | 0.0 | - | - | - | 0.0 | 7.5 | - | - | 100.0 | n/a | 3785 | 1.9 | 0.0238 |
| M2 | llm_only:structured | case57 | 40 | 0.0 | - | - | - | 0.0 | 5.3 | 0.0 | 0.0 | 100.0 | n/a | 7056 | 1.5 | 0.0413 |
| M2 | llm_only:structured | case118 | 40 | 0.0 | - | - | - | 0.0 | 7.5 | - | - | 100.0 | n/a | 16254 | 2.8 | 0.0986 |
| M2 | single_call:cot | case14 | 40 | 90.0 | 90.0 | 1.54e-05 | 0.05 | 100.0 | 99.2 | - | - | 0.0 | 1.9 | 9631 | 3.3 | 0.0614 |
| M2 | single_call:cot | case30 | 40 | 90.0 | 90.0 | 1.80e-05 | 0.04 | 100.0 | 99.6 | - | - | 0.0 | 1.9 | 14797 | 3.8 | 0.0923 |
| M2 | single_call:cot | case57 | 40 | 90.0 | 90.0 | 1.34e-05 | 0.10 | 100.0 | 96.7 | 0.0 | 0.0 | 15.8 | 1.9 | 29979 | 7.6 | 0.1822 |
| M2 | single_call:cot | case118 | 40 | 95.0 | 95.0 | 3.26e-06 | 1.44 | 100.0 | 100.0 | - | - | 0.0 | 2.0 | 56013 | 6.0 | 0.3401 |
| M2 | single_call:few_shot | case14 | 40 | 95.0 | 95.0 | 1.47e-05 | 0.05 | 100.0 | 98.5 | - | - | 0.0 | 1.9 | 10017 | 3.4 | 0.0634 |
| M2 | single_call:few_shot | case30 | 40 | 92.5 | 92.5 | 1.76e-05 | 0.04 | 100.0 | 99.6 | - | - | 0.0 | 2.0 | 15190 | 3.8 | 0.0946 |
| M2 | single_call:few_shot | case57 | 40 | 90.0 | 90.0 | 1.34e-05 | 0.10 | 100.0 | 99.0 | 0.0 | 0.0 | 13.2 | 1.9 | 28610 | 4.6 | 0.1670 |
| M2 | single_call:few_shot | case118 | 40 | 92.5 | 92.5 | 6.98e-06 | 1.44 | 100.0 | 100.0 | - | - | 0.0 | 2.0 | 55239 | 6.3 | 0.3354 |
| M2 | single_call:rag | case14 | 40 | 95.0 | 95.0 | 1.47e-05 | 0.05 | 100.0 | 99.0 | - | - | 0.0 | 1.7 | 7546 | 4.2 | 0.0485 |
| M2 | single_call:rag | case30 | 40 | 90.0 | 90.0 | 1.76e-05 | 0.04 | 100.0 | 99.5 | - | - | 2.5 | 1.8 | 10028 | 4.0 | 0.0634 |
| M2 | single_call:rag | case57 | 40 | 90.0 | 90.0 | 1.34e-05 | 0.10 | 100.0 | 99.5 | 0.0 | 0.0 | 13.2 | 1.7 | 19868 | 5.1 | 0.1169 |
| M2 | single_call:rag | case118 | 40 | 95.0 | 95.0 | 3.26e-06 | 1.44 | 100.0 | 99.6 | - | - | 0.0 | 1.8 | 31823 | 5.9 | 0.1942 |
| M2 | single_call:structured | case14 | 40 | 95.0 | 95.0 | 1.47e-05 | 0.05 | 100.0 | 99.1 | - | - | 0.0 | 2.0 | 10175 | 4.0 | 0.0644 |
| M2 | single_call:structured | case30 | 40 | 92.5 | 92.5 | 1.76e-05 | 0.04 | 100.0 | 99.5 | - | - | 0.0 | 2.0 | 15058 | 4.6 | 0.0939 |
| M2 | single_call:structured | case57 | 40 | 90.0 | 90.0 | 1.34e-05 | 0.10 | 100.0 | 98.2 | 0.0 | 0.0 | 13.2 | 1.9 | 29396 | 6.0 | 0.1716 |
| M2 | single_call:structured | case118 | 40 | 92.5 | 92.5 | 6.98e-06 | 1.44 | 100.0 | 100.0 | - | - | 0.0 | 2.0 | 56340 | 7.3 | 0.3420 |
| M2 | react_nogate | case14 | 40 | 97.5 | 97.5 | 1.90e-07 | 0.02 | 100.0 | 98.2 | - | - | 0.0 | 2.3 | 7620 | 4.2 | 0.0494 |
| M2 | react_nogate | case30 | 40 | 97.5 | 97.5 | 1.43e-06 | 0.02 | 100.0 | 100.0 | - | - | 0.0 | 2.3 | 10048 | 4.1 | 0.0642 |
| M2 | react_nogate | case57 | 40 | 92.5 | 92.5 | 0 | 0.05 | 100.0 | 97.2 | 0.0 | 0.0 | 10.5 | 2.2 | 18230 | 5.1 | 0.1085 |
| M2 | react_nogate | case118 | 40 | 100.0 | 100.0 | 0 | 1.33 | 100.0 | 99.5 | - | - | 0.0 | 2.3 | 29816 | 5.2 | 0.1833 |
| M2 | react | case14 | 40 | 97.5 | 97.5 | 1.90e-07 | 0.02 | 100.0 | 98.7 | - | - | 0.0 | 2.3 | 7641 | 4.1 | 0.0495 |
| M2 | react | case30 | 40 | 100.0 | 100.0 | 0 | 0.00 | 100.0 | 100.0 | - | - | 0.0 | 2.3 | 10113 | 4.2 | 0.0646 |
| M2 | react | case57 | 40 | 92.5 | 92.5 | 0 | 0.05 | 100.0 | 97.2 | 0.0 | 0.0 | 10.5 | 2.2 | 18769 | 5.1 | 0.1113 |
| M2 | react | case118 | 40 | 100.0 | 100.0 | 0 | 1.33 | 100.0 | 99.5 | - | - | 0.0 | 2.4 | 32834 | 5.7 | 0.2013 |
| M2 | plan_act_nogate | case14 | 40 | 47.5 | 47.5 | 4.36e-04 | 0.07 | 100.0 | 98.5 | - | - | 2.5 | 2.8 | 5738 | 4.2 | 0.0382 |
| M2 | plan_act_nogate | case30 | 40 | 42.5 | 42.5 | 2.65e-04 | 0.05 | 100.0 | 100.0 | - | - | 2.5 | 2.8 | 7916 | 4.3 | 0.0512 |
| M2 | plan_act_nogate | case57 | 40 | 42.5 | 42.5 | 3.96e-04 | 0.11 | 100.0 | 99.2 | 0.0 | 0.0 | 0.0 | 2.6 | 15913 | 4.9 | 0.0948 |
| M2 | plan_act_nogate | case118 | 40 | 45.0 | 45.0 | 3.65e-05 | 1.44 | 100.0 | 100.0 | - | - | 0.0 | 2.8 | 27040 | 5.0 | 0.1665 |
| M2 | plan_act | case14 | 40 | 47.5 | 47.5 | 4.36e-04 | 0.07 | 100.0 | 98.5 | - | - | 0.0 | 2.8 | 5736 | 4.7 | 0.0382 |
| M2 | plan_act | case30 | 40 | 42.5 | 42.5 | 2.65e-04 | 0.05 | 100.0 | 100.0 | - | - | 2.5 | 2.8 | 7914 | 4.3 | 0.0512 |

| Model | Method | Case | N | Solved | Form. | V_MAE | F_MAER | Faith. | SFR | Claim | Abst. | Calls | Tok. | TTV | \$ | |
|-------|----------|---------|----|--------|-------|----------|--------|--------|-------|-------|-------|-------|------|-------|-----|--------|
| M2 | plan_act | case57 | 40 | 42.5 | 42.5 | 3.96e-04 | 0.11 | 100.0 | 98.8 | 0.0 | 0.0 | 0.0 | 2.6 | 15918 | 5.3 | 0.0949 |
| M2 | plan_act | case118 | 40 | 45.0 | 45.0 | 3.65e-05 | 1.44 | 100.0 | 100.0 | – | – | 0.0 | 2.8 | 27043 | 5.3 | 0.1666 |
| M2 | pfagent | case14 | 40 | 97.5 | 97.5 | 1.90e-07 | 0.02 | 100.0 | 98.7 | – | – | 0.0 | 2.3 | 7511 | 4.0 | 0.0488 |
| M2 | pfagent | case30 | 40 | 95.0 | 95.0 | 8.63e-05 | 0.04 | 100.0 | 100.0 | – | – | 0.0 | 2.2 | 9235 | 4.0 | 0.0593 |
| M2 | pfagent | case57 | 40 | 92.5 | 92.5 | 0 | 0.05 | 100.0 | 97.4 | 0.0 | 0.0 | 7.9 | 2.2 | 18779 | 4.9 | 0.1115 |
| M2 | pfagent | case118 | 40 | 100.0 | 100.0 | 0 | 1.33 | 100.0 | 100.0 | – | – | 0.0 | 2.4 | 32365 | 5.4 | 0.1986 |

Agregado por método (media ponderada por items sobre todos los casos)

Modelos: M1 = none:rule_based; M2 = openrouter:openai/gpt-4o-mini

| Model | Method | Case | N | Solved | Form. | V_MAE | F_MAER | Faith. | SFR | Claim | Abst. | Calls | Tok. | TTV | \$ | |
|-------|------------------------|---------|-----|--------|-------|----------|--------|--------|-------|-------|-------|-------|------|-------|-----|--------|
| M1 | rule_based | 4 casos | 160 | 67.5 | 67.5 | 2.82e-05 | 0.42 | 100.0 | 100.0 | 0.0 | 0.0 | 26.2 | 1.8 | n/a | 0.1 | 0.0000 |
| M2 | llm_only:cot | 4 casos | 160 | 0.0 | – | – | – | 0.0 | 15.1 | 0.0 | 0.0 | 98.8 | n/a | 8004 | 5.7 | 0.2251 |
| M2 | llm_only:few_shot | 4 casos | 160 | 0.0 | – | – | – | 0.0 | 6.3 | 0.0 | 0.0 | 98.8 | n/a | 7898 | 2.0 | 0.1917 |
| M2 | llm_only:rag | 4 casos | 160 | 0.0 | – | – | – | 0.0 | 6.3 | 0.0 | 0.0 | 98.8 | n/a | 841 | 1.8 | 0.0243 |
| M2 | llm_only:structured | 4 casos | 160 | 0.0 | – | – | – | 0.0 | 6.3 | 0.0 | 0.0 | 98.8 | n/a | 7394 | 2.0 | 0.1797 |
| M2 | single_call:cot | 4 casos | 160 | 91.2 | 91.2 | 1.24e-05 | 0.42 | 100.0 | 98.9 | 0.0 | 0.0 | 3.8 | 1.9 | 27605 | 5.2 | 0.6761 |
| M2 | single_call:few_shot | 4 casos | 160 | 92.5 | 92.5 | 1.32e-05 | 0.41 | 100.0 | 99.3 | 0.0 | 0.0 | 3.1 | 2.0 | 27264 | 4.5 | 0.6605 |
| M2 | single_call:rag | 4 casos | 160 | 92.5 | 92.5 | 1.22e-05 | 0.41 | 100.0 | 99.4 | 0.0 | 0.0 | 3.8 | 1.7 | 17316 | 4.8 | 0.4229 |
| M2 | single_call:structured | 4 casos | 160 | 92.5 | 92.5 | 1.32e-05 | 0.41 | 100.0 | 99.2 | 0.0 | 0.0 | 3.1 | 2.0 | 27742 | 5.5 | 0.6719 |
| M2 | react_nogate | 4 casos | 160 | 96.9 | 96.9 | 4.09e-07 | 0.36 | 100.0 | 98.7 | 0.0 | 0.0 | 2.5 | 2.3 | 16428 | 4.6 | 0.4053 |
| M2 | react | 4 casos | 160 | 97.5 | 97.5 | 4.80e-08 | 0.35 | 100.0 | 98.9 | 0.0 | 0.0 | 2.5 | 2.3 | 17339 | 4.8 | 0.4267 |
| M2 | plan_act_nogate | 4 casos | 160 | 44.4 | 44.4 | 2.82e-04 | 0.42 | 100.0 | 99.4 | 0.0 | 0.0 | 1.2 | 2.7 | 14151 | 4.6 | 0.3507 |
| M2 | plan_act | 4 casos | 160 | 44.4 | 44.4 | 2.82e-04 | 0.42 | 100.0 | 99.3 | 0.0 | 0.0 | 0.6 | 2.7 | 14153 | 4.9 | 0.3508 |
| M2 | pfagent | 4 casos | 160 | 96.2 | 96.2 | 2.19e-05 | 0.36 | 100.0 | 99.0 | 0.0 | 0.0 | 1.9 | 2.3 | 16973 | 4.6 | 0.4183 |

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

| Method | plain | parameterized | multistep | ambiguous |
|------------------------|---------------|---------------|--------------|--------------|
| llm_only:cot | n/a | n/a | n/a | n/a |
| llm_only:few_shot | n/a | n/a | n/a | n/a |
| llm_only:rag | n/a | n/a | n/a | n/a |
| llm_only:structured | n/a | n/a | n/a | n/a |
| single_call:cot | 100.0 (40/40) | 97.5 (39/40) | 90.0 (36/40) | 77.5 (31/40) |
| single_call:few_shot | 100.0 (40/40) | 97.5 (39/40) | 97.5 (39/40) | 75.0 (30/40) |
| single_call:rag | 100.0 (40/40) | 97.5 (39/40) | 95.0 (38/40) | 77.5 (31/40) |
| single_call:structured | 100.0 (40/40) | 97.5 (39/40) | 97.5 (39/40) | 75.0 (30/40) |
| rule_based | 80.0 (32/40) | 57.5 (23/40) | 75.0 (30/40) | 57.5 (23/40) |
| react_nogate | 100.0 (40/40) | 97.5 (39/40) | 95.0 (38/40) | 95.0 (38/40) |
| react | 100.0 (40/40) | 97.5 (39/40) | 95.0 (38/40) | 97.5 (39/40) |
| plan_act_nogate | 60.0 (24/40) | 47.5 (19/40) | 32.5 (13/40) | 37.5 (15/40) |
| plan_act | 60.0 (24/40) | 47.5 (19/40) | 32.5 (13/40) | 37.5 (15/40) |
| pfagent | 100.0 (40/40) | 97.5 (39/40) | 92.5 (37/40) | 95.0 (38/40) |

Solved por dificultad (% , items resueltos / items)

| Method | plain | parameterized | multistep | ambiguous |
|------------------------|---------------|---------------|--------------|--------------|
| llm_only:cot | 0.0 (0/40) | 0.0 (0/40) | 0.0 (0/40) | 0.0 (0/40) |
| llm_only:few_shot | 0.0 (0/40) | 0.0 (0/40) | 0.0 (0/40) | 0.0 (0/40) |
| llm_only:rag | 0.0 (0/40) | 0.0 (0/40) | 0.0 (0/40) | 0.0 (0/40) |
| llm_only:structured | 0.0 (0/40) | 0.0 (0/40) | 0.0 (0/40) | 0.0 (0/40) |
| single_call:cot | 100.0 (40/40) | 97.5 (39/40) | 90.0 (36/40) | 77.5 (31/40) |
| single_call:few_shot | 100.0 (40/40) | 97.5 (39/40) | 97.5 (39/40) | 75.0 (30/40) |
| single_call:rag | 100.0 (40/40) | 97.5 (39/40) | 95.0 (38/40) | 77.5 (31/40) |
| single_call:structured | 100.0 (40/40) | 97.5 (39/40) | 97.5 (39/40) | 75.0 (30/40) |
| rule_based | 80.0 (32/40) | 57.5 (23/40) | 75.0 (30/40) | 57.5 (23/40) |
| react_nogate | 100.0 (40/40) | 97.5 (39/40) | 95.0 (38/40) | 95.0 (38/40) |
| react | 100.0 (40/40) | 97.5 (39/40) | 95.0 (38/40) | 97.5 (39/40) |
| plan_act_nogate | 60.0 (24/40) | 47.5 (19/40) | 32.5 (13/40) | 37.5 (15/40) |
| plan_act | 60.0 (24/40) | 47.5 (19/40) | 32.5 (13/40) | 37.5 (15/40) |
| pfagent | 100.0 (40/40) | 97.5 (39/40) | 92.5 (37/40) | 95.0 (38/40) |

Tipos de error de formulación (conteo de items)

| Method | extra_arg_changes_result | invalid_value | missed_step | ok | unparsed | wrong_id |
|----------------------|--------------------------|---------------|-------------|-----|----------|----------|
| single_call:cot | 0 | 0 | 3 | 146 | 2 | 9 |
| single_call:few_shot | 1 | 0 | 0 | 148 | 2 | 9 |
| single_call:rag | 0 | 0 | 1 | 148 | 2 | 9 |

| Method | extra_arg_changes_result | invalid_value | missed_step | ok | unparsed | wrong_id |
|------------------------|--------------------------|---------------|-------------|-----|----------|----------|
| single_call:structured | 1 | 0 | 0 | 148 | 2 | 9 |
| rule_based | 0 | 0 | 0 | 108 | 41 | 11 |
| react_nogate | 0 | 0 | 1 | 155 | 2 | 2 |
| react | 0 | 0 | 1 | 156 | 2 | 1 |
| plan_act_nogate | 52 | 26 | 0 | 71 | 2 | 9 |
| plan_act | 52 | 26 | 0 | 71 | 2 | 9 |
| pfagent | 0 | 0 | 2 | 154 | 2 | 2 |

Errores de ejecución por método (conteo de items con error)

| Method | GroundTruthNotConverged | formulation_failure | no_solver_result |
|------------------------|-------------------------|---------------------|------------------|
| llm_only:cot | 2 | 0 | 0 |
| llm_only:few_shot | 2 | 0 | 0 |
| llm_only:rag | 2 | 0 | 0 |
| llm_only:structured | 2 | 0 | 0 |
| single_call:cot | 2 | 0 | 3 |
| single_call:few_shot | 2 | 0 | 0 |
| single_call:rag | 2 | 0 | 0 |
| single_call:structured | 2 | 0 | 0 |
| rule_based | 2 | 39 | 0 |
| react_nogate | 2 | 0 | 0 |
| react | 2 | 0 | 0 |
| plan_act_nogate | 2 | 0 | 0 |
| plan_act | 2 | 0 | 0 |
| pfagent | 2 | 0 | 0 |

4. Trazas de ejemplo

Hasta 2 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método llm_only:cot

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
1. The request refers to the following buses:
  - Buses: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14
  - Slack bus: 0 (external grid)
  - Load buses: 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13
  - Generator buses: 1, 2, 5, 7

2. The line involved in the request is:
  - From bus 10 to bus 11 (line_id 12)

3. The quantities and units requested are:
  - Voltage magnitude (p.u.) and phase angle (degrees) for each node
  - Active power (MW) and load factor (%) at the beginning of each branch
  - Total generation (MW), total load (MW), total losses (MW)
  - Voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)

4. Disconnecting the line between bus 10 and bus 11 will affect the power flow in the network. The power balance must hold, meaning that
↳ the total generation must equal the ...[552 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=2994; cost_usd=0.0006; wall=3.6 s
```

Ejemplo: no resuelto

```
request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
↳ line between bus 6 and bus 13 (1-based).
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
1. The request refers to the following buses:
  - Slack bus: Bus 1 (External grid)
  - Load buses: Buses 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13
  - Generator buses: Buses 1, 2, 5, 7
```

2. The line being disconnected is between bus 6 and bus 13, which corresponds to line_id 14 in the line data.

3. The quantities requested include:

- Voltage magnitude (p.u.) and phase angle (degrees) for each node
- Active power (MW) and load factor (%) at the beginning of each branch
- Total generation (MW), total load (MW), and total losses (MW)
- Voltage overruns (<0.95 or >1.05 p.u.) and line overruns (>100%)

4. Disconnecting the line between bus 6 and bus 13 will affect the power flow in the network. The power balance must hold, meaning that the
↳ total generation must equal the total load plu ...[402 chars más]

--- Scoring ---

formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=2940; cost_usd=0.0006; wall=5.0 s

Método llm_only:few_shot

Ejemplo: con bandera de fallo (claimed/stale/abstained)

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})

executed_calls: (sin etapa de tools)

fuelle de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)

--- Respuesta final ---

```
{
  "converged": false,
  "bus_voltages": [],
  "line_flows": [],
  "total_generation_mw": 0.0,
  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

--- Scoring ---

formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=2967; cost_usd=0.0005; wall=1.3 s

Ejemplo: no resuelto

request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
↳ line between bus 6 and bus 13 (1-based).

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})

executed_calls: (sin etapa de tools)

fuelle de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)

--- Respuesta final ---

```
{
  "converged": false,
  "bus_voltages": [],
  "line_flows": [],
  "total_generation_mw": 0.0,
  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

--- Scoring ---

formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=2981; cost_usd=0.0005; wall=1.3 s

Método llm_only:rag

Ejemplo: con bandera de fallo (claimed/stale/abstained)

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})

executed_calls: (sin etapa de tools)

fuelle de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)

--- Respuesta final ---

```
{
  "converged": false,
  "bus_voltages": [],
  "line_flows": [],
  "total_generation_mw": 0.0,
  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

--- Scoring ---

formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=802; cost_usd=0.0001; wall=1.4 s

Ejemplo: no resuelto

```

request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
↳ line between bus 6 and bus 13 (1-based).
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{
  "converged": false,
  "bus_voltages": [],
  "line_flows": [],
  "total_generation_mw": 0.0,
  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=908; cost_usd=0.0002; wall=1.6 s

```

Método llm_only:structured

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{
  "converged": false,
  "bus_voltages": [],
  "line_flows": [],
  "total_generation_mw": 0.0,
  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=2463; cost_usd=0.0004; wall=2.2 s

```

Ejemplo: no resuelto

```

request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
↳ line between bus 6 and bus 13 (1-based).
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{
  "converged": false,
  "bus_voltages": [],
  "line_flows": [],
  "total_generation_mw": 0.0,
  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=2478; cost_usd=0.0004; wall=1.9 s

```

Método single_call:cot

Ejemplo: resuelto (solved=True)

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: disconnect_line({"from_bus": 10, "to_bus": 11}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 4, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 5, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 6, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 7, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 8, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 9, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 10, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 11, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 12, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 13, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 14, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}], "line_flows": [{"line_id": 1, "from_bus": 1, "to_bus": 2, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 2, "from_bus": 2, "to_bus": 3, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 3, "from_bus": 3, "to_bus": 4, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 4, "from_bus": 4, "to_bus": 5, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 5, "from_bus": 5, "to_bus": 6, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 6, "from_bus": 6, "to_bus": 7, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 7, "from_bus": 7, "to_bus": 8, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 8, "from_bus": 8, "to_bus": 9, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 9, "from_bus": 9, "to_bus": 10, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 10, "from_bus": 10, "to_bus": 11, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 11, "from_bus": 11, "to_bus": 12, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 12, "from_bus": 12, "to_bus": 13, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 13, "from_bus": 13, "to_bus": 14, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 14, "from_bus": 14, "to_bus": 1, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}], "total_generation_mw": 0.0, "total_load_mw": 0.0, "total_loss_mw": 0.0}
  gate: PASS (reasons=, mismatch=7.85e-09 MW)
tool: run_powerflow({})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 4, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 5, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 6, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 7, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 8, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 9, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 10, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 11, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 12, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 13, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 14, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}], "line_flows": [{"line_id": 1, "from_bus": 1, "to_bus": 2, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 2, "from_bus": 2, "to_bus": 3, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 3, "from_bus": 3, "to_bus": 4, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 4, "from_bus": 4, "to_bus": 5, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 5, "from_bus": 5, "to_bus": 6, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 6, "from_bus": 6, "to_bus": 7, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 7, "from_bus": 7, "to_bus": 8, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 8, "from_bus": 8, "to_bus": 9, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 9, "from_bus": 9, "to_bus": 10, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 10, "from_bus": 10, "to_bus": 11, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 11, "from_bus": 11, "to_bus": 12, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 12, "from_bus": 12, "to_bus": 13, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 13, "from_bus": 13, "to_bus": 14, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}, {"line_id": 14, "from_bus": 14, "to_bus": 1, "p_mw": 0.0, "q_mvar": 0.0, "status": "OK"}], "total_generation_mw": 0.0, "total_load_mw": 0.0, "total_loss_mw": 0.0}
  gate: PASS (reasons=, mismatch=7.85e-09 MW)
--- Ronda 2 [final] ---
LLM: The line between bus 10 and bus 11 has been successfully disconnected, and the power flow has been resolved.

```

Key numbers:
- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.01 pu (bus 3) to 1.09 pu (bus 8)

Violations:
- Overvoltage violations at buses ...[154 chars más]
--- Respuesta final ---
The line between bus 10 and bus 11 has been successfully disconnected, and the power flow has been resolved.

Key numbers:
- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.01 pu (bus 3) to 1.09 pu (bus 8)

Violations:
- Overvoltage violations at buses 1, 6, 7, 8, 11, and 12.

Suggested next step: Consider running a contingency analysis or recommending remedial actions to address the voltage violations.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=9777;
↪ cost_usd=0.0015; wall=3.3 s

Ejemplo: formulación fallida (formulation_exact=False)

request (case14-ambiguous-027-s0, case14, difficulty=ambiguous, seed=203647507): Load the IEEE 14-bus system and set the active load at bus
↪ 10 (0-based) to 4.7 MW.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 11, "p_mw": 4.7})
executed_calls: modify_load({"bus_id": 10, "p_mw": 4.7}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: modify_load({"bus_id": 10, "p_mw": 4.7})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.9130978627475494, "is_violation": false,
↪ "violation_type": null}, {"bus_id": 3, "vm_pu": ...[6449 chars más]
gate: PASS (reasons=-, mismatch=4.77e-09 MW)
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.9130978627475494, "is_violation": false,
↪ "violation_type": null}, {"bus_id": 3, "vm_pu": ...[6449 chars más]
gate: PASS (reasons=-, mismatch=4.77e-09 MW)
--- Ronda 2 [final] ---
LLM: The load at bus 10 has been successfully modified to 4.7 MW, and the power flow has converged.

- Total load: 260.02 MW
- Total generation: 273.56 MW
- Total losses: 13.54 MW
- Voltage range: 1.0369 pu to 1.0900 pu

Violations:
- Overvoltage violations are present at buses 1, 6, 7, 8, 9, 10, 11, 12, ...[138 chars más]
--- Respuesta final ---
The load at bus 10 has been successfully modified to 4.7 MW, and the power flow has converged.

- Total load: 260.02 MW
- Total generation: 273.56 MW
- Total losses: 13.54 MW
- Voltage range: 1.0369 pu to 1.0900 pu

Violations:
- Overvoltage violations are present at buses 1, 6, 7, 8, 9, 10, 11, 12, and 13.

Suggested next step: Consider running a contingency analysis or recommending remedial actions to address the voltage violations.
--- Scoring ---
formulation=wrong_id (exact=False); V_MAE=5.79e-04; F_MAE=1.22; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↪ tokens=10210; cost_usd=0.0016; wall=3.2 s

Método single_call:few_shot

Ejemplo: resuelto (solved=True)

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↪ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: disconnect_line({"from_bus": 10, "to_bus": 11}) -> get_most_loaded_branch({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": ...[5930 chars más]
gate: PASS (reasons=-, mismatch=7.85e-09 MW)
tool: get_most_loaded_branch({})

```

    output: {"branch_id": 0, "branch_type": "line", "from_bus": 1, "to_bus": 2, "loading_percent": 1.4487892295883087}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: The line between bus 10 and bus 11 has been successfully disconnected.

```

Key numbers:

- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.06 pu (bus 1) to 1.09 pu (bus 8)

Violations:

- Overvoltage violations at buses 1, 6, 7, 8, 11, and 12.

Suggested n ...[117 chars más]

--- Respuesta final ---

The line between bus 10 and bus 11 has been successfully disconnected.

Key numbers:

- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.06 pu (bus 1) to 1.09 pu (bus 8)

Violations:

- Overvoltage violations at buses 1, 6, 7, 8, 11, and 12.

Suggested next step: Consider running a contingency analysis or recommending remedial actions to address the voltage violations.

--- Scoring ---

```

formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=7491;
↳ cost_usd=0.0012; wall=4.3 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-ambiguous-027-s0, case14, difficulty=ambiguous, seed=203647507): Load the IEEE 14-bus system and set the active load at bus
↳ 10 (0-based) to 4.7 MW.

```

```

intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 11, "p_mw": 4.7})

```

```

executed_calls: modify_load({"bus_id": 10, "p_mw": 4.7}) -> run_powerflow({})

```

fuentes de la traza: archivo de traza completo

--- Ronda 1 ---

```

tool: modify_load({"bus_id": 10, "p_mw": 4.7})

```

```

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.9130978627475494, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_ ...[6449 chars más]
gate: PASS (reasons=-, mismatch=4.77e-09 MW)

```

```

tool: run_powerflow({})

```

```

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.9130978627475494, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_ ...[6449 chars más]
gate: PASS (reasons=-, mismatch=4.77e-09 MW)

```

--- Ronda 2 [final] ---

LLM: The load at bus 10 has been successfully modified to 4.7 MW.

- Total load: 260.02 MW
- Total generation: 273.56 MW
- Total losses: 13.54 MW
- Voltage range: 1.0369 pu to 1.0900 pu

There are multiple voltage violations at buses 1, 6, 7, 8, 9, 10, 11, 12, and 13, all experiencing overvoltage issues ...[131 chars más]

--- Respuesta final ---

The load at bus 10 has been successfully modified to 4.7 MW.

- Total load: 260.02 MW
- Total generation: 273.56 MW
- Total losses: 13.54 MW
- Voltage range: 1.0369 pu to 1.0900 pu

There are multiple voltage violations at buses 1, 6, 7, 8, 9, 10, 11, 12, and 13, all experiencing overvoltage issues.

Suggested next step: Consider running a contingency analysis or recommending remedial actions to address the voltage violations.

--- Scoring ---

```

formulation=wrong_id (exact=False); V_MAE=5.79e-04; F_MAE=1.22; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=10414; cost_usd=0.0016; wall=4.0 s

```

Método single_call:rag

Ejemplo: resuelto (solved=True)

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.

```

```

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})

```

```

executed_calls: disconnect_line({"from_bus": 10, "to_bus": 11})

```

fuentes de la traza: archivo de traza completo

--- Ronda 1 ---

```

tool: disconnect_line({"from_bus": 10, "to_bus": 11})

```

```

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 4, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 5, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 6, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 7, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 8, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 9, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 10, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 11, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 12, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}], "gate": PASS (reasons=, mismatch=7.85e-09 MW)
--- Ronda 2 [final] ---
LLM: The line between bus 10 and bus 11 has been successfully disconnected.

Key numbers: total load 253.85 MW, total generation 266.51 MW, total losses 12.66 MW, voltage range includes violations at several buses.

Violations: Overvoltage violations are present at buses 1, 6, 7, 8, 11, and 12.

Suggest ...[91 chars más]
--- Respuesta final ---
The line between bus 10 and bus 11 has been successfully disconnected.

Key numbers: total load 253.85 MW, total generation 266.51 MW, total losses 12.66 MW, voltage range includes violations at several buses.

Violations: Overvoltage violations are present at buses 1, 6, 7, 8, 11, and 12.

Suggested next step: Consider implementing remedial actions to address the overvoltage violations.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4961;
  ↳ cost_usd=0.0008; wall=3.7 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-ambiguous-027-s0, case14, difficulty=ambiguous, seed=203647507): Load the IEEE 14-bus system and set the active load at bus
  ↳ 10 (0-based) to 4.7 MW.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 11, "p_mw": 4.7})
executed_calls: modify_load({"bus_id": 10, "p_mw": 4.7})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: modify_load({"bus_id": 10, "p_mw": 4.7})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 4, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 5, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 6, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 7, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 8, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 9, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 10, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 11, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 12, "vm_pu": 1.01, "va_deg": -4.9130978627475494, "is_violation": false,
  ↳ "violation_type": null}], "gate": PASS (reasons=, mismatch=4.77e-09 MW)
--- Ronda 2 [final] ---
LLM: The load at bus 10 has been successfully modified to 4.7 MW, and the system has converged.

- Total load: 260.02 MW
- Total generation: 273.56 MW
- Total losses: 13.54 MW
- Voltage range: 1.06 pu (bus 1) to 1.0546 pu (bus 10)

Violations: There are multiple overvoltage violations at buses 1, 6, 7, ...[164 chars más]
--- Respuesta final ---
The load at bus 10 has been successfully modified to 4.7 MW, and the system has converged.

- Total load: 260.02 MW
- Total generation: 273.56 MW
- Total losses: 13.54 MW
- Voltage range: 1.06 pu (bus 1) to 1.0546 pu (bus 10)

Violations: There are multiple overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, 12, and 13.

Suggested next step: Consider running a contingency analysis or generating remedial-action suggestions to address the voltage violations.
--- Scoring ---
formulation=wrong_id (exact=False); V_MAE=5.79e-04; F_MAE=1.22; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
  ↳ tokens=5155; cost_usd=0.0008; wall=5.0 s

```

Método single_call:structured

Ejemplo: resuelto (solved=True)

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
  ↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: disconnect_line({"from_bus": 10, "to_bus": 11}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 4, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 5, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 6, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 7, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 8, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 9, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 10, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 11, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 12, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}], "gate": PASS (reasons=, mismatch=7.85e-09 MW)
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 4, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 5, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 6, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 7, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 8, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 9, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 10, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 11, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 12, "vm_pu": 1.01, "va_deg": -4.782856812869876, "is_violation": false,
  ↳ "violation_type": null}], "gate": PASS (reasons=, mismatch=7.85e-09 MW)
--- Ronda 2 [final] ---
LLM: The line between bus 10 and bus 11 has been successfully disconnected.

Key numbers:
- Total load: 253.85 MW

```

- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.01 pu to 1.09 pu

Violations:

- Overvoltage violations at buses 1, 6, 7, 8, 11, and 12.

Suggested next step: Consid ...[101 chars más]

--- Respuesta final ---

The line between bus 10 and bus 11 has been successfully disconnected.

Key numbers:

- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.01 pu to 1.09 pu

Violations:

- Overvoltage violations at buses 1, 6, 7, 8, 11, and 12.

Suggested next step: Consider running a contingency analysis or recommending remedial actions to address the voltage violations.

--- Scoring ---

formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=9606;

↪ cost_usd=0.0015; wall=4.0 s

Ejemplo: formulación fallida (formulation_exact=False)

request (case14-ambiguous-027-s0, case14, difficulty=ambiguous, seed=203647507): Load the IEEE 14-bus system and set the active load at bus 10 (0-based) to 4.7 MW.

intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 11, "p_mw": 4.7})

executed_calls: modify_load({"bus_id": 10, "p_mw": 4.7}) -> run_powerflow({})

fuelle de la traza: archivo de traza completo

--- Ronda 1 ---

tool: modify_load({"bus_id": 10, "p_mw": 4.7})

```
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.9130978627475494, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_ ...[6449 chars más]
gate: PASS (reasons=-, mismatch=4.77e-09 MW)
```

tool: run_powerflow({})

```
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.9130978627475494, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_ ...[6449 chars más]
gate: PASS (reasons=-, mismatch=4.77e-09 MW)
```

--- Ronda 2 [final] ---

LLM: The load at bus 10 has been successfully modified to 4.7 MW, and the power flow has converged.

- Total load: 260.02 MW
- Total generation: 273.56 MW
- Total losses: 13.54 MW
- Voltage range: 1.0369 pu to 1.0900 pu

Violations:

- Overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, 12, and 13.

Su ...[97 chars más]

--- Respuesta final ---

The load at bus 10 has been successfully modified to 4.7 MW, and the power flow has converged.

- Total load: 260.02 MW
- Total generation: 273.56 MW
- Total losses: 13.54 MW
- Voltage range: 1.0369 pu to 1.0900 pu

Violations:

- Overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, 12, and 13.

Suggested next step: Consider running a remedial action analysis to address the voltage violations.

--- Scoring ---

formulation=wrong_id (exact=False); V_MAE=5.79e-04; F_MAE=1.22; faithful=100.0 %; solved=False (formulation); flags=-; error=-;

↪ tokens=10049; cost_usd=0.0016; wall=5.6 s

Método rule_based

Ejemplo: resuelto (solved=True)

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and bus eleven.

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})

executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})

fuelle de la traza: archivo de traza completo

--- Ronda 1 ---

tool: load_case({"case_name": "case14"})

```
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
  ↪ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
```

tool: disconnect_line({"from_bus": 10, "to_bus": 11})


```

**Violations:** There are multiple overvoltage violations at buses 1, ...[166 chars más]
--- Respuesta final ---
The active load at bus 1 has been set to 11 MW, and the power flow has been successfully solved.

- **Total Load:** 263.83 MW
- **Total Generation:** 276.50 MW
- **Total Losses:** 12.67 MW
- **Voltage Range:** 1.035 to 1.090 p.u.

**Violations:** There are multiple overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, and 12.

**Suggested Next Step:** Consider running a contingency analysis or recommending remedial actions to address the overvoltage violations.
--- Scoring ---
formulation=wrong_id (exact=False); V_MAE=7.59e-06; F_MAE=0.61; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=5362; cost_usd=0.0009; wall=4.0 s

```

Método react

Ejemplo: resuelto (solved=True)

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -5930 chars más}
  gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 2 ---
LLM: The case14 has been loaded, and the line between bus 10 and bus 11 has been successfully disconnected.

- **Key numbers**: Total load: 253.85 MW, Total generation: 266.51 MW, Total losses: 12.66 MW, Voltage range: 1.01 to 1.07 p.u.
- **Violations**: There are multiple overvoltage violations at buses ...[150 chars más]
--- Respuesta final ---
The case14 has been loaded, and the line between bus 10 and bus 11 has been successfully disconnected.

- **Key numbers**: Total load: 253.85 MW, Total generation: 266.51 MW, Total losses: 12.66 MW, Voltage range: 1.01 to 1.07 p.u.
- **Violations**: There are multiple overvoltage violations at buses 1, 6, 7, 8, 11, and 12.
- **Suggested next step**: Consider running a contingency analysis to assess the impact of this disconnection on the network.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4353;
↳ cost_usd=0.0007; wall=3.1 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-ambiguous-039-s0, case14, difficulty=ambiguous, seed=315597205): Load the 14-bus test case (case14) and set the active load
↳ at bus 1 (0-based) to 11 MW.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 2, "p_mw": 11.0})
executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 1, "p_mw": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.83309137785733,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: modify_load({"bus_id": 1, "p_mw": 11})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.823083955414936, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0099999 ...[6289 chars más}
  gate: PASS (reasons=-, mismatch=6.18e-09 MW)
--- Ronda 3 ---
LLM: The active load at bus 1 has been set to 11 MW, and the power flow has been successfully solved.

- **Total Load**: 263.83 MW
- **Total Generation**: 276.50 MW
- **Total Losses**: 12.67 MW
- **Voltage Range**: 1.06 to 1.09 p.u.

**Violations**: There are multiple overvoltage violations at buses 1, 6 ...[173 chars más]
--- Respuesta final ---
The active load at bus 1 has been set to 11 MW, and the power flow has been successfully solved.

- **Total Load**: 263.83 MW
- **Total Generation**: 276.50 MW
- **Total Losses**: 12.67 MW
- **Voltage Range**: 1.06 to 1.09 p.u.

**Violations**: There are multiple overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, and 12.

```

```

**Suggested Next Step**: Consider running a contingency analysis or generating remedial action suggestions to address the overvoltage
↳ violations.
--- Scoring ---
formulation=wrong_id (exact=False); V_MAE=7.59e-06; F_MAE=0.61; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=5369; cost_usd=0.0009; wall=4.0 s

```

Método plan_act_nogate

Ejemplo: resuelto (solved=True)

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 [plan] ---
LLM: {"plan":[{"tool":"load_case","args":{"case_name":"case14"}},{"tool":"disconnect_line","args":{"from_bus":10,"to_bus":11}}]}
--- Ronda 2 [act] ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va ...[5930 chars más]
  gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 3 [answer] [final] ---
LLM: The power flow analysis has been completed after disconnecting the line between bus 10 and bus 11.

1. Summary: The system is converged, but there are multiple voltage violations.
2. Key numbers: Total load = 253.85 MW, Total generation = 266.51 MW, Total losses = 12.66 MW, Voltage range = 1.01 to 1 ...[216 chars más]
--- Respuesta final ---
The power flow analysis has been completed after disconnecting the line between bus 10 and bus 11.

```

```

1. Summary: The system is converged, but there are multiple voltage violations.
2. Key numbers: Total load = 253.85 MW, Total generation = 266.51 MW, Total losses = 12.66 MW, Voltage range = 1.01 to 1.09 p.u.
3. Violations: There are overvoltage violations at buses 1, 6, 7, 8, 11, and 12.
4. Suggested next step: Consider implementing voltage control measures or load shedding to address the overvoltage violations.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=3613;
↳ cost_usd=0.0006; wall=4.5 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-ambiguous-011-s0, case14, difficulty=ambiguous, seed=1971893210): Load case14 and set the active load at bus 6 to 14,700 kw.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 6, "p_mw": 14.7})
executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 0})
fuente de la traza: archivo de traza completo
--- Ronda 1 [plan] ---
LLM: {"plan":[{"tool":"load_case","args":{"case_name":"case14"}},{"tool":"modify_load","args":{"bus_id":6,"p_mw":14.7,"q_mvar":0}}]}
--- Ronda 2 [act] ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 263.13580543670497,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
tool: modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 0})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.044999999999997, "va_deg": -5.236323374534588, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[6297 chars más]
  gate: PASS (reasons=-, mismatch=6.47e-09 MW)
--- Ronda 3 [answer] [final] ---
LLM: The power flow analysis has been completed successfully after modifying the load at bus 6.

1. Summary: The system is converged, but there are multiple voltage violations.
2. Key numbers: Total load = 267.15 MW, Total generation = 281.82 MW, Total losses = 14.67 MW, Voltage range = 1.0353 to 1.0900 ...[233 chars
↳ más]
--- Respuesta final ---
The power flow analysis has been completed successfully after modifying the load at bus 6.

```

```

1. Summary: The system is converged, but there are multiple voltage violations.
2. Key numbers: Total load = 267.15 MW, Total generation = 281.82 MW, Total losses = 14.67 MW, Voltage range = 1.0353 to 1.0900 p.u.
3. Violations: There are overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, and 12.
4. Suggested next step: Consider running a contingency analysis or generating remedial-action suggestions to address the voltage violations.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=9.52e-17; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=-;
↳ error=-; tokens=3792; cost_usd=0.0007; wall=3.1 s

```

Método plan_act

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 [plan] ---
LLM: {"plan":[{"tool":"load_case","args":{"case_name":"case14"}},{tool":"disconnect_line","args":{"from_bus":10,"to_bus":11}}]}
--- Ronda 2 [act] ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5930 chars más}
gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 3 [answer] [final] ---
LLM: The power flow analysis has been completed after disconnecting the line between bus 10 and bus 11.

1. Summary: The system is converged, but there are multiple voltage violations.
2. Key numbers: Total load = 253.85 MW, Total generation = 266.51 MW, Total losses = 12.66 MW, Voltage range = 1.01 to 1 ...[206 chars más]
--- Respuesta final ---
The power flow analysis has been completed after disconnecting the line between bus 10 and bus 11.
```

```
1. Summary: The system is converged, but there are multiple voltage violations.
2. Key numbers: Total load = 253.85 MW, Total generation = 266.51 MW, Total losses = 12.66 MW, Voltage range = 1.01 to 1.09 p.u.
3. Violations: Overvoltage violations at buses 1, 6, 7, 8, 11, and 12.
4. Suggested next step: Consider implementing voltage control measures or load shedding to address the overvoltage violations.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=3610;
↳ cost_usd=0.0006; wall=3.1 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-011-s0, case14, difficulty=ambiguous, seed=1971893210): Load case14 and set the active load at bus 6 to 14,700 kw.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 6, "p_mw": 14.7})
executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 0})
fuente de la traza: archivo de traza completo
--- Ronda 1 [plan] ---
LLM: {"plan":[{"tool":"load_case","args":{"case_name":"case14"}},{tool":"modify_load","args":{"bus_id":6,"p_mw":14.7,"q_mvar":0}}]}
--- Ronda 2 [act] ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 263.13580543670497,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 0})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.236323374534588, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 6297 chars más}
gate: PASS (reasons=-, mismatch=6.47e-09 MW)
--- Ronda 3 [answer] [final] ---
LLM: The power flow analysis has been completed successfully after modifying the load at bus 6.

1. Summary: The system is converged but has multiple voltage violations.
2. Key numbers: Total load = 267.15 MW, Total generation = 281.82 MW, Total losses = 14.67 MW, Voltage range = 1.0353 to 1.0900 p.u.
3. ...[212 chars más]
--- Respuesta final ---
The power flow analysis has been completed successfully after modifying the load at bus 6.
```

```
1. Summary: The system is converged but has multiple voltage violations.
2. Key numbers: Total load = 267.15 MW, Total generation = 281.82 MW, Total losses = 14.67 MW, Voltage range = 1.0353 to 1.0900 p.u.
3. Violations: There are overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, and 12.
4. Suggested next step: Consider implementing voltage control measures or load shedding to address the overvoltage violations.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=9.52e-17; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=-;
↳ error=-; tokens=3790; cost_usd=0.0007; wall=4.0 s
```

Método pfagent

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5930 chars más}
```

```

gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 2 ---
LLM: The case14 has been loaded, and the line between bus 10 and bus 11 has been successfully disconnected.

- **Key numbers**: Total load: 253.85 MW, Total generation: 266.51 MW, Total losses: 12.66 MW, Voltage range: 1.01 to 1.09 p.u.
- **Violations**: There are several overvoltage violations at buses ...[161 chars más]
--- Respuesta final ---
The case14 has been loaded, and the line between bus 10 and bus 11 has been successfully disconnected.

- **Key numbers**: Total load: 253.85 MW, Total generation: 266.51 MW, Total losses: 12.66 MW, Voltage range: 1.01 to 1.09 p.u.
- **Violations**: There are several overvoltage violations at buses 1, 6, 7, 8, 11, and 12.
- **Suggested next step**: Consider running a contingency analysis to assess the impact of this disconnection on the network's stability.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4355;
↳ cost_usd=0.0007; wall=3.6 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-ambiguous-039-s0, case14, difficulty=ambiguous, seed=315597205): Load the 14-bus test case (case14) and set the active load
↳ at bus 1 (0-based) to 11 MW.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 2, "p_mw": 11.0})
executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 1, "p_mw": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.83309137785733,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: modify_load({"bus_id": 1, "p_mw": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.823083955414936, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0099999 ...[6289 chars más]
gate: PASS (reasons=-, mismatch=6.18e-09 MW)
--- Ronda 3 ---
LLM: The active load at bus 1 has been set to 11 MW, and the power flow has been successfully solved.

- **Total Load**: 263.83 MW
- **Total Generation**: 276.50 MW
- **Total Losses**: 12.67 MW
- **Voltage Range**: 1.06 to 1.09 p.u.

**Violations**: There are multiple overvoltage violations at buses 1, 6 ...[135 chars más]
--- Respuesta final ---
The active load at bus 1 has been set to 11 MW, and the power flow has been successfully solved.

- **Total Load**: 263.83 MW
- **Total Generation**: 276.50 MW
- **Total Losses**: 12.67 MW
- **Voltage Range**: 1.06 to 1.09 p.u.

**Violations**: There are multiple overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, and 12.

**Suggested Next Step**: Consider running a remedial action analysis to address the overvoltage violations.
--- Scoring ---
formulation=wrong_id (exact=False); V_MAE=7.59e-06; F_MAE=0.61; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=5365; cost_usd=0.0009; wall=3.8 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- $V_MAE \geq 1e-4$ p.u. en 6 filas método×caso con solver: plan_act/case14 (4.36e-04), plan_act_nogate/case14 (4.36e-04), plan_act/case30 (2.65e-04), plan_act_nogate/case30 (2.65e-04), plan_act/case57 (3.96e-04), plan_act_nogate/case57 (3.96e-04).
- LLM-only: ningún método×caso tiene V_MAE (ningún item devolvió tensiones comparables).
- 98.8 % (632/640) de abstención en LLM-only (respuesta con converged=false o arrays vacíos).
- En LLM-only, 632 items no resueltos por convergence_mismatch y 0 por voltage_error.
- El gate verificó 2050 resultados de flujo, 523 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- 39 items con formulation_failure: el método no formuló ninguna tool call.
- 28 items con GroundTruthNotConverged: la verdad de referencia no convergió y el item no puntúa.
- 3 items con error no_solver_result.
- Solved más alto: react 97.5 % (156/160); más bajo: llm_only:cot 0.0 % (0/160).
- Formulación exacta agregada sobre métodos con tools: 81.6 % (1305/1600).
- Formulación por dificultad: mejor plain 90.0 % (360/400), peor ambiguous 72.5 % (290/400).
- Errores de formulación más frecuentes: extra_arg_changes_result=106, wrong_id=70, unparsed=59.
- Ningún item reclamó éxito sobre un fallo (claimed_success_on_failure=0).
- De 28 items en estado de fallo, 0 lo declararon de forma segura (SFR 0.0 % (0/28)).

- 35 de 993 items con mutación de red reportaron estado obsoleto (stale_state).
- Trazabilidad de números (Faith.): 99.2% con tools frente a 8.5% en LLM-only.
- Costo total 5.0039 USD; el método más caro fue `single_call:cot` con 0.6761 USD (14%).
- Métodos con tools: 2.17 tool calls medias por item (máximo 6).
- Solved por modelo: `none:rule_based` 67.5% (108/160); `openrouter:openai/gpt-4o-mini` 57.5% (1197/2080).

6. Archivos

Rutas absolutas bajo `results_matrix_gpt-4o-mini.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_matrix_gpt-4o-mini/llm_only_cot/case14/report.json [leído:
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scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
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